

Flexible supercapacitor for wearables and EVs

Researchers at Nagaland University (Lumami, Nagaland) have developed a flexible supercapacitor that combines



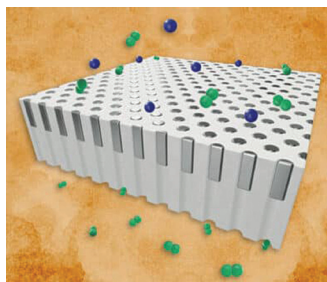
Nagaland University scientists develop flexible supercapacitor for wearables and electric vehicles (Credit: <https://indiaeducationdiary.in>)

high energy density, mechanical flexibility, and durability, addressing significant limits of traditional batteries in wearables, EVs, and IoT devices. Using cobalt-doped molybdenum diselenide, the device achieves $34.54 \text{ W} \cdot \text{h} \cdot \text{kg}^{-1}$ and over 10,000 stable charge cycles, even

under bending. Eco-friendly synthesis and a working prototype highlight practical viability, paving the way for fast-charging, bendable power sources in next-generation electronics.

Separating hydrogen from other gases at high temperature made easier

MIT (Cambridge, Massachusetts) engineers have developed a breakthrough palladium membrane that replaces continu-



Schematic illustration of the membrane showing selective permeation of hydrogen (green) from a mixture of hydrogen and helium (blue) gases (Photo Credit: Lohyun Kim)

ous films with palladium plugs embedded in a porous support. Hydrogen production has long struggled with separating hydrogen from other gases at high temperatures, as traditional palladium membranes degrade above 800K. These plugs stay stable up to 1000K, maintaining selectivity without degradation. The design

reduces reactor size, cost, and energy use, advancing hydrogen generation in processes like ammonia cracking and steam methane reforming.

The first 2D-silicon hybrid chip

Researchers at Fudan University in Shanghai have unveiled the world's first memory chip that fuses atom-thin 2D materials with conventional silicon circuitry. Led by Chunsen Liu at Fudan University, the team used their ATOM2CHIP method to grow ultra-thin memory cells directly on silicon, achieving stable electrical contact and robust packaging. The fully operational chip, tested at 5MHz, delivers faster performance with lower energy use, promises a new era of efficient, high-speed computing.

A finger mouse with long battery life

University of Tokyo researchers have unveiled picoRing, a five-gram, ultra-low-power ring mouse that turns a finger



picoRing mouse

30 - 500 uW

5 gram

The picoRing is a finger mouse with a long battery life (Credit: <https://www.hackster.io>)

into a precise control device for computers and VR systems. Unlike bulky wearables, picoRing delivers 600-1000 hours of operation on a 27mAh charge by using inductive telemetry instead of Bluetooth, reducing power consumption to microwatts. Its miniature trackball and resonant coil enable smooth, always-on interaction, paving the way for invisible, sensor-rich wearable interfaces of the future.

Flying robots that swap tools midair

Researchers at Westlake University, China, have developed FlyingToolbox, a system that lets drones exchange tools midair using QR-based vision, neural downwash correction,



The FlyingToolbox support drone system (Credit: <https://interestingengineering.com>)

and magnetic docking. Industrial inspection, maintenance, and construction often involve dangerous or hard-to-reach areas, but drones with robotic arms are making these tasks safer and more efficient. Their main

drawback has been poor midair cooperation due to turbulence. This breakthrough enables continuous, coordinated, and flexible aerial operations in hazardous environments.

A new layer on solar panels makes them more efficient and durable

Scientists at the University of New South Wales (Kensington, Australia) have developed a new carbon-based DPND layer that boosts solar panel efficiency to around 30%, up from today's 20-25%, while keeping panels about 2.4°C cooler. Built on a process called singlet fission, this organic layer captures high-energy light and converts it into extra power instead of heat, improving durability, lowering costs, and extending panel lifespan by over four years.